Expectation Response to Portfolio Gains

1. 2026.8.23

1.1 Data sources

- **Behavior and trading tables.**
  - `invest` (positions and monthly returns)
  - `trade` (transaction records)
  - `fund` (fund master)
  - `csmar` (third-party fund NAV for cross-validation)
  - `sme_business` (firm-level activity for behavior outcomes).
- **Survey tables.**
  - Five waves of `smesurvey`: 2024q2, 2024q3, 2024q4, 2025q1, 2025q2.[1]
- **User attributes.**
  - `sme_portrait` supplies user-level controls (risk preference, age, region, etc.).

1.2 Indicators

- **X / Z (instrument).**
  - `passive_ret_month`: quasi-lottery IV. Up to now, we use the return within a natural month, without adjusting for the submit date. Xiaohan: as in my note, we can use CSMAR daily data.
- **X (endogenous).**
  - `realized_ret_month`: realized monthly return on the platform books.
  - Xiaohan: let's use reduced-form for non-end-of-month variables.
- **Y, four-wave macro.**
  - `exp_gdp`, `exp_cpi`, `exp_house`, `exp_stock`, `exp_rate`.

[1] The 2024q2 employer segment is missing macro expectation columns. The current analysis locks to the employer segment, so 2024q2 macro outcomes are simply absent.

- **Y, five-wave micro.**
  - `exp_env_local`, `exp_rev`, `exp_market`, `exp_price`, `exp_wage`, `exp_input_cost`.
- **Controls.**
  - `portfolio_m_end`, `n_fund_m_end`.
  - Xiaohan: Discuss next time. I tried several in my notes.

1.3 V-codes mapping across waves

Table 1.1. V-code inventory across the five waves (11 expectations). Macro columns are blank for employer in 2024q2. [verifed.]

Y	2024q2	2024q3	2024q4	2025q1	2025q2
`exp_stock`	—	v196	v187	v181	v190
`exp_gdp`	—	v192	v182	v176	v185
`exp_cpi`	—	v195	v186	v180	v189
`exp_house`	—	v193	v184	v178	v187
`exp_rate`	—	v194	v185	v179	v188
`exp_env_local`	v203	v212	v203	v198	v206
`exp_rev`	v104	v107	v93	v94	v107
`exp_market`	v103	v106	v92	v93	v106
`exp_price`	v176	v189	v179	v173	v182
`exp_wage`	v177	v190	v180	v174	v183
`exp_input_cost`	v178	v191	v181	v175	v184

Table 1.2. Question descriptions for the 11 expectation concepts.

Y	问题简述 (Chinese description)	跨 wave Q 题号
`exp_stock`	未来 12 月上证指数	Q56 / Q53 / Q56 / Q57
`exp_gdp`	未来 12 月 GDP 增速	Q52 / Q48 / Q51 / Q52
`exp_cpi`	未来 12 月 CPI	Q55 / Q52 / Q55 / Q56
`exp_house`	整体房价增速	Q53 / Q50 / Q53 / Q54
`exp_rate`	5 年期贷款利率	Q54 / Q51 / Q54 / Q55
`exp_env_local`	当地经营环境	Q56 / Q60 / Q57 / Q61 / Q61
`exp_rev`	营业收入 / 销售额 (预期下季)	Q45 / Q47 / Q45 / Q46 / Q49
`exp_market`	市场需求 / 接受新订单 (预期)	Q45 / Q47 / Q45 / Q46 / Q49
`exp_price`	主要产品价格 (预期下季)	Q47 / Q50 / Q47 / Q50 / Q51
`exp_wage`	全职员工工资 (预期下季)	Q47 / Q50 / Q47 / Q50 / Q51
`exp_input_cost`	原材料价格 (预期下季)	Q47 / Q50 / Q47 / Q50 / Q51

1.4 Mapping dictionaries

`PCT_DICT` (used by nine Y's: `exp_gdp`, `exp_cpi`, `exp_house`, `exp_rate`, `exp_market`, `exp_price`, `exp_wage`, `exp_input_cost`, `exp_rev`). All values are in percent units; positive = upward change, negative = downward change. Open-ended bracket midpoints are anchored at a representative point (e.g. 下降 3% 以上 = −4).

Table 1.3. `PCT_DICT` Part 1.

Positive bin	Raw text (positive)	Value (%)	Negative bin	Raw text (negative)	Value (%)
$[+20\%, +\infty)$	增长 20_ 以上	+25	—	基本不变 / 没有明显变化	0
$[+10\%, +20\%)$	增长 20_ 以内	+10	$[-10\%, 0)$	降低 10_ 以内	−5
$[+10\%, +\infty)$	增长 10_ 以上	+15	$(-\infty, -10\%]$	降低 10_ 以上	−15
$[0, +10\%)$	增长 10_ 以内	+5	$[-20\%, 0)$	降低 20_ 以内	−10
			$(-\infty, -20\%]$	降低 20_ 以上	−25

Table 1.4. `PCT_DICT` Part 2. **A** = `exp_gdp` (8 coarse bins, GDP 4% midpoint for the 3–4% bin); **B** = `exp_cpi` (9 fine bins, half-point resolution around zero); **C** = `exp_house` / `exp_rate` (5 asymmetric bins using 上涨/降低 wording)[房价应该是 D?; 利率应该是每 1pt 一个区间?]; **D** [should it be Table 1.3 instead? The three price indicators shall be C? Also note that 2024q1 and 2024q2 can be slightly different.]= `exp_market` / `exp_price` / `exp_wage` / `exp_input_cost` / `exp_rev` (10 symmetric bins covering wider magnitudes, 上升/下降 wording).

[Update: Yanpeng has a large mapping table in code instead of these. Will check.]

Scale	Bin	Raw text	Value (%)
A	$(-\infty, -3\%]$	下降 3% 以上	−4.0
	$[-3\%, 0]$	下降 0–3%	−1.5
	$[0, +3\%)$	增长 0–3%	+1.5
	$[+3\%, +4\%)$	增长 3–4%	+3.5
	$[+4\%, +5\%)$	增长 4–5%	+4.5
	$[+5\%, +6\%)$	增长 5–6%	+5.5
	$[+6\%, +8\%)$	增长 6–8%	+7.0
	$[+8\%, +\infty)$	增长 8% 以上	+9.0
B	$(-\infty, -5\%]$	下降 5% 以上	−6.0
	$[-5\%, -3\%)$	下降 3–5%	−4.0
	$[-3\%, -2\%)$	下降 2–3%	−2.5
	$[-2\%, -1\%)$	下降 1–2%	−1.5
	$[-1\%, 0)$	下降 0–1%	−0.5
	$[0, +1\%)$	增长 0–1%	+0.5
	$[+1\%, +2\%)$	增长 1–2%	+1.5
	$[+2\%, +3\%)$	增长 2–3%	+2.5
	$[+3\%, +\infty)$	增长 3% 以上	+4.0
C	$(-\infty, -5\%]$	降低 3–5%	−4.0
	$[-3\%, 0)$	降低 3% 以内	−2.0
	$[0, +3\%)$	上涨 3% 以内	+2.0
	$[+3\%, +5\%)$	上涨 3–5%	+4.0
	$[+5\%, +\infty)$	上涨 5% 以上	+6.0
D	$(-\infty, -30\%]$	下降 30% 以上	−35.0
	$[-30\%, -15\%]$	下降 15–30%	−22.5
	$[-15\%, -10\%]$	下降 10–15%	−12.5
	$[-10\%, -5\%]$	下降 5–10%	−7.5
	$[-5\%, 0)$	下降 0–5%	−2.5
	$[0, +5\%)$	上升 0–5%	+2.5
	$[+5\%, +10\%)$	上升 5–10%	+7.5
	$[+10\%, +15\%)$	上升 10–15%	+12.5
	$[+15\%, +30\%)$	上升 15–30%	+22.5
	$[+30\%, +\infty)$	上升 30% 以上	+35.0

Two notes on A / B reuse: the literal raw-text string 增长 `0_3` is mapped to +1.5 and shared by both A and B; and the half-open $[0, +3\%)$ cell of A has the same midpoint as the B $[0, +1\%)$ cell after rounding to one decimal (+1.5). Both are intentional.

Table 1.5. `PCT_DICT` Part 3. All five uncodable strings below collapse to NaN and are dropped at regression time.

Raw text	Value	Meaning
不清楚	NaN	respondent does not know
不确定	NaN	respondent is unsure
不适用	NaN	question does not apply to this respondent
关心但拿不准	NaN	respondent cares but cannot judge

Table 1.6. `INDEX_DICT` (used only by `exp_stock`; index-range midpoint in absolute points).

Bin (index points)	Midpoint	Bin (index points)	Midpoint	Bin (index points)	Midpoint
$(-\infty, 2200)$	2000.0	$[2600, 2800)$	2700.0	$[3600, 3800)$	3700.0
$[2200, 2400)$	2300.0	$[2800, 3000)$	2900.0	$[3800, 4000)$	3900.0
$(-\infty, 2600)$	2400.0	$[3000, 3300)$	3150.0	$[3600, +\infty)$	3800.0
$[2400, 2600)$	2500.0	$[3300, 3600)$	3450.0	$[4000, +\infty)$	4200.0
不确定	NaN	不适用	NaN		

Table 1.7. `DIRECTION_DICT` (used only by `exp_env_local`). Two equivalent wordings (bare direction vs. 预期-prefixed) are merged to the same numeric code.

Raw text	Value	Raw text	Value
恶化	−1	预期好转	+1
预期变差	−1	好转	+1
不变	0	不清楚	NaN
预期不变	0	不确定	NaN
		不适用	NaN

All three dictionaries route uncodable answers to NaN, so each regression cleanly excludes them.

1.5 Derived indicators

Definitions of the indicators that are computed rather than read directly from the platform.

- **`passive_ret_month` (X / IV).** At the user-month level, defined as [2]

$$\sum_{f} (\text{hold}_{u,f,t-1} \times \text{fund_ret}_{f,t}) \,/\, \text{portfolio_m_prev},$$

where the numerator is the sum across all funds the user held at the end of month $t-1$ of the per-fund return in month t, and the denominator is the previous month's total market

[2] Up to now we use the return within a natrual month, without adjusting for the submit date.

value of the portfolio (in the dataset, portfolio_m_prev is just the sum of $\text{hold}_{u,f,t-1}$, right?). The numerator is the dollar return the user would have earned without trading; dividing by the previous portfolio normalizes it to a ratio in $[-1, 1]$, comparable across users of different portfolio sizes.

- **`realized_ret_month` (X / endogenous).** At the user-month level, defined as the sum across all funds of the platform's 当月累计月收益元 column from the `invest` table. This is what the platform actually booked into the user's account.
- **`portfolio_m_prev`.** Total market value of the portfolio at the end of month $t-1$. Serves as the denominator of `passive_ret_month`.

1.6 Distribution findings

Findings from the diagnostic phase that pin the data structure behind the chosen spec. [Can you present the reduced-form regression table?]

- **Survey raw_value stability across waves.** `exp_stock` exhibits a stable frequency structure across its four available waves, supporting pooled 2SLS estimation. `exp_cpi` is similarly stable. `exp_gdp` shows somewhat more drift across waves. `exp_market`, with five-wave coverage, is clean enough for direct mapping. Update: This is for an old version. But can still show a reduced form/IV/OLS.
- **Portrait distributions.** 风险等级 (risk preference) is concentrated, usable as a continuous control. [定义：-2 已过期，-1 未做过，0-5 越大越激进；0 极度保守型,1 保守型,2 稳健型,3 平衡型,4 成长型,5 积极型] 年龄 has a small outlier tail (n = 270 outside $[18, 80]$), captured by `age_outlier`. `tenure` is right-skewed; the same closed-form definition applies across waves.
- **No-variation controls excluded.** 是否商户 (98.22%) and 是否收钱码 (96.34%) have almost no variation in the sample and are therefore not introduced as controls.

Selected key distributions. All cross-wave frequency values are percentage points (*pp*) of wave respondents; the (缺失) row aggregates non-respondents plus respondents whose raw value fell outside the scale (e.g., 2024q2 employee-segment macro columns). Empty cells denote a raw value that did not appear in that wave.

1.6.1 Macro Expectations.

[better to present within the asked subsample.]

Table 1.8. `exp_stock` raw_value distribution across the four available waves.

raw_value	2024q3 (%)	2024q4 (%)	2025q1 (%)	2025q2 (%)
2200_2400 点	1.31			
2200 点以下	1.60			
2400_2600 点	1.30			
2600_2800 点	1.71	1.09	1.80	1.74
2600 点以下		1.50	1.58	1.68
2800_3000 点	2.79	1.44	1.81	2.01
3000_3300 点	3.59	2.34	2.65	3.81
3300_3600 点	2.75	2.29	3.69	3.69
3600_3800 点		1.46	2.44	2.27
3600 点以上	3.32			
3800_4000 点		1.45	2.16	1.88
4000 点以上		2.18	2.66	2.12
不确定	12.24	9.90	10.48	10.90
(缺失)	69.39	76.35	70.73	69.91

Table 1.9. `exp_cpi` raw_value distribution across the four available waves.

raw_value	2024q3 (%)	2024q4 (%)	2025q1 (%)	2025q2 (%)
下降 0_1	2.28	1.98	2.65	2.90
下降 1_2	2.89	2.25	2.82	3.15
下降 2_3	1.92	1.71	1.97	2.29
下降 3_5	1.47	1.18	1.25	1.52
下降 5_ 以上	2.13	1.59	1.83	1.90
不确定	9.03	6.83	8.19	8.05
增长 0_1	2.93	2.25	3.19	3.43
增长 1_2	3.45	2.64	3.79	3.49
增长 2_3	2.19	1.54	1.81	1.76
增长 3_ 以上	2.33	1.70	1.78	1.61
(缺失)	69.39	76.35	70.73	69.91

Table 1.10. `exp_gdp` raw_value distribution across the four available waves.

raw_value	2024q3 (%)	2024q4 (%)	2025q1 (%)	2025q2 (%)
下降 0_3	2.18	1.57	1.87	2.41
下降 3_ 以上	2.50	1.87	1.94	2.20
不确定	10.44	8.35	9.55	9.66
增长 0_3	3.63	3.06	3.74	4.72
增长 3_4	3.55	2.44	3.52	3.57
增长 4_5	2.94	2.17	3.55	2.85
增长 5_6	2.30	1.45	1.97	1.98
增长 6_8	1.36	1.04	1.38	1.22
增长 8_ 以上	1.73	1.70	1.76	1.49
(缺失)	69.39	76.35	70.73	69.91

Table 1.11. `exp_house` raw_value distribution across the four available waves. Values are percentage points (*pp*) of wave respondents. Empty cells denote a raw value that did not appear in that wave.

raw_value	2024q3 (%)	2024q4 (%)	2025q1 (%)	2025q2 (%)
上升 0_5	2.49	2.08	2.72	2.36
上升 5_10	1.25	1.30	1.41	1.25
上升 10_15	1.08	0.78	1.15	0.93
上升 15_30	0.93	0.74	0.82	0.70
上升 30_ 以上	1.07	0.80	0.95	0.78
下降 0_5	3.81	2.45	3.84	4.17
下降 5_10	3.25	2.53	3.64	4.14
下降 10_15	2.52	2.02	2.28	2.59
下降 15_30	2.34	1.54	1.91	2.33
下降 30_ 以上	2.11	1.84	1.83	1.90
不确定	9.77	7.58	8.72	8.97
(缺失)	69.39	76.35	70.73	69.91

Table 1.12. `exp_rate` raw_value distribution across the four available waves. Values are percentage points (*pp*) of wave respondents. The `exp_rate` scale uses integer bins (0–1%, 1–2%, ..., 5% 以上) rather than half-percent cells. Empty cells denote a raw value that did not appear in that wave.

raw_value	2024q3 (%)	2024q4 (%)	2025q1 (%)	2025q2 (%)
0_1	2.93	2.38	2.82	4.11
1_2	4.68	3.42	4.93	5.61
2_3	6.32	4.87	6.33	6.06
3_4	4.46	3.34	4.12	3.50
4_5	2.10	1.68	1.90	2.07
5_ 以上	2.10	1.52	1.98	1.73
不确定	8.02	6.44	7.21	7.02
(缺失)	69.39	76.35	70.73	69.91

1.6.2 Micro Expectations.

Table 1.13. `exp_market` raw_value distribution across all five waves.

raw_value	2024q2 (%)	2024q3 (%)	2024q4 (%)	2025q1 (%)	2025q2 (%)
不清楚	12.25	12.08	12.66	11.16	11.83
基本不变	22.90	19.62	20.52	21.17	24.28
增长 10_ 以上	3.40	7.12	7.29	6.49	6.05
增长 10_ 以内	3.60	6.78	7.85	7.16	6.91
增长 20_ 以上	3.42				
增长 20_ 以内	4.11				
降低 10_ 以上	3.29	6.75	8.20	6.54	6.37
降低 10_ 以内	2.90	5.46	7.48	6.27	6.76
降低 20_ 以上	2.99				
降低 20_ 以内	4.03				
(缺失)	37.12	42.18	35.99	41.21	37.81

Table 1.14. `exp_env_local` raw_value distribution across all five waves. Values are percentage points (*pp*) of wave respondents. Only three raw values appear in this scale (预期不变 / 预期变差 / 预期好转); the (缺失) row is noticeably larger than the four-wave expectations because the employer segment is missing in 2024q2.

raw_value	2024q2 (%)	2024q3 (%)	2024q4 (%)	2025q1 (%)	2025q2 (%)
预期不变	9.88	10.22	8.49	12.98	13.32
预期变差	6.59	5.80	4.87	5.55	5.43
预期好转	14.33	14.60	10.29	10.74	11.34
(缺失)	69.21	69.39	76.35	70.73	69.91

Table 1.15. `exp_rev` raw_value distribution across all five waves. Values are percentage points (*pp*) of wave respondents. The 2024q3–2025q2 columns reuse the CONFI general ordinal; the 2024q2 column carries the same labels but with smaller percentages for the 增长 20/降低 20 bins, reflecting the early-wave CONFI wording.

raw_value	2024q2 (%)	2024q3 (%)	2024q4 (%)	2025q1 (%)	2025q2 (%)
不清楚	10.33	10.68	11.11	9.82	10.28
基本不变	23.32	19.48	20.95	21.14	24.15
增长 10_ 以上	3.47	7.19	7.58	6.71	6.38
增长 10_ 以内	3.96	7.35	8.60	7.87	7.70
增长 20_ 以上	3.25				
增长 20_ 以内	4.82				
降低 10_ 以上	3.54	7.04	7.90	6.37	6.38
降低 10_ 以内	3.30	6.08	7.87	6.88	7.30
降低 20_ 以上	2.64				
降低 20_ 以内	4.26				
(缺失)	37.12	42.18	35.99	41.21	37.81

Table 1.16. `exp_price` raw_value distribution across all five waves. Values are percentage points (*pp*) of wave respondents. The 2024q2 column carries the early-wave wording (上涨 3_5 / 上涨 3_ 以内 / 上涨 5_ 以上 / 上涨 5_ 以内 with sparse / 不关心 / 关心但拿不准); the 2024q3–2025q2 columns use the standard 上涨/降低 3-bin scale.

raw_value	2024q2 (%)	2024q3 (%)	2024q4 (%)	2025q1 (%)	2025q2 (%)
上涨 3_5		2.47	1.60	2.19	1.87
上涨 3_ 以内		4.16	2.85	3.73	3.98
上涨 5_ 以上	3.41	2.51	1.83	2.05	2.28
上涨 5_ 以内	4.91				
不关心	2.35				
不确定		5.85	5.57	5.64	5.48
关心但拿不准	4.20				
基本不变	12.62	11.79	8.95	11.55	12.32
降低 3_5		1.06	0.87	1.39	1.23
降低 3_ 以内		1.50	0.92	1.55	1.90
降低 5_ 以上	1.45	1.27	1.06	1.17	1.03
降低 5_ 以内	1.85				
(缺失)	69.21	69.39	76.35	70.73	69.91

Table 1.17. `exp_wage` raw_value distribution across all five waves. Values are percentage points (*pp*) of wave respondents. Identical wording pattern as `exp_price`.

raw_value	2024q2 (%)	2024q3 (%)	2024q4 (%)	2025q1 (%)	2025q2 (%)
上涨 3_5		1.67	1.17	1.70	1.64
上涨 3_ 以内		2.26	1.79	2.37	2.48
上涨 5_ 以上	2.68	2.36	1.95	2.29	2.11
上涨 5_ 以内	2.62				
不关心	2.49				
不确定		6.31	5.80	6.41	6.21
关心但拿不准	5.51				
基本不变	12.73	12.12	8.26	10.98	12.02
降低 3_5		1.75	1.19	1.51	1.36
降低 3_ 以内		1.66	1.42	1.60	1.91
降低 5_ 以上	2.59	2.50	2.08	2.42	2.38
降低 5_ 以内	2.17				
(缺失)	69.21	69.39	76.35	70.73	69.91

Table 1.18. `exp_input_cost` raw_value distribution across all five waves. Values are percentage points (*pp*) of wave respondents. Identical wording pattern as `exp_price` and `exp_wage`.

raw_value	2024q2 (%)	2024q3 (%)	2024q4 (%)	2025q1 (%)	2025q2 (%)
上涨 3_5		1.13	0.86	1.24	1.36
上涨 3_ 以内		1.77	1.53	1.79	1.94
上涨 5_ 以上	1.76	1.50	1.00	1.50	1.40
上涨 5_ 以内	2.24				
不关心	3.42				
不确定		5.28	5.01	4.97	4.92
关心但拿不准	2.32				
基本不变	18.22	17.30	12.26	15.88	16.74
降低 3_5		1.09	0.90	1.06	0.97
降低 3_ 以内		1.26	0.94	1.31	1.60
降低 5_ 以上	1.43	1.30	1.14	1.52	1.18
降低 5_ 以内	1.40				
(缺失)	69.21	69.39	76.35	70.73	69.91

1.6.3 Portrait.

Table 1.19. portrait primary-key sanity. Confirms 匿名化用户 `id` uniqueness and join compatibility with survey / invest.

Table	行数	唯一用户数	重复数
`[portrait]`	42,791	42,791	0

Table 1.20. portrait 年龄 describe. 42,522 respondents age in [18,80].

stat	value	stat	value
n_total	42,791	n_null (raw)	69
n_valid	42,722	min	7.00
p01	19.00	p05	21.00
p25	29.00	median	36.00
p75	45.00	p95	60.00
p99	72.00	max	102.00
mean	38.02	std	12.17

Table 1.21. portrait 性别 value_counts. Two valid values (1, 2) plus 39 observations with value 0.

性别	count	pct
1	27,154	63.5%
2	15,598	36.5%
0	39	0.1%
(缺失)	0	0.0%

Table 1.22. portrait 是否商户 value_counts. Indicator equal to 1 for the 98.22% of respondents that are merchants.

是否商户	count	pct
1	42,031	98.2%
0	760	1.8%
(缺失)	0	0.0%

Table 1.23. portrait 常驻省份 value_counts. Top 15 provinces plus 其他; the top three (广东 / 浙江 / 江苏) account for 35.0% of respondents.

常驻省份	count	pct
广东省	7,817	18.3%
浙江省	3,788	8.9%
江苏省	3,328	7.8%
河南省	3,167	7.4%
山东省	2,872	6.7%
福建省	2,021	4.7%
湖北省	2,002	4.7%
上海	1,739	4.1%
安徽省	1,647	3.8%
河北省	1,566	3.7%
江西省	1,410	3.3%
湖南省	1,351	3.2%
广西壮族自治区	1,302	3.0%
四川省	1,237	2.9%
山西省	1,124	2.6%
其他	6,207	14.5%
(缺失)	213	0.5%

Table 1.24. portrait 常驻城市 value_counts. Top 15 cities plus 其他. [I suggest use the city-tier category variable.]

常驻城市	count	pct
上海市	1,739	4.1%
广州市	1,592	3.7%
深圳市	1,346	3.1%
杭州市	1,106	2.6%
武汉市	917	2.1%
东莞市	896	2.1%
北京市	844	2.0%
郑州市	777	1.8%
苏州市	741	1.7%
南京市	631	1.5%
佛山市	620	1.4%
成都市	613	1.4%
重庆市	546	1.3%
温州市	539	1.3%
汕头市	536	1.3%
其他	29,135	68.1%
(缺失)	213	0.5%

Table 1.25. portrait 商户类别 value_counts. Top 15 categories plus 其他; 其他服务 dominates at 81.4% because the merchant category classifier defaults to that bucket when no other category is recorded. [details were asked in the questionnaire. To be updated.]

商户类别	count	pct
其他服务	34,816	81.4%
其他实物	3,168	7.4%
餐饮服务	672	1.6%
汽车类	616	1.4%
粮油、食品、饮料、烟酒类	439	1.0%
互联网服务	430	1.0%
文化娱乐服务	411	1.0%
日用品类	408	1.0%
生活服务	202	0.5%
物流服务	169	0.4%
家用电器和音像器材类	136	0.3%
服装鞋帽、针、纺织品类	129	0.3%
建筑及装潢材料类	120	0.3%
文化办公用品类	50	0.1%
化妆品类	43	0.1%
其他	221	0.5%
(缺失)	761	1.8%

Let's list the summary for 风险等级 as well.

1.7 Lessons learned

- **Fund-month panel must be reindexed before any `shift(1)` lag.** A naive `shift(1)` on the raw `invest` panel misaligns at scale: at least 27,220 fund-months jump by exactly one month and 15,397 jump by two or more. The lag must therefore be applied after reindexing the (`uid`, `fund`) panel to a full monthly grid with missing months filled as zero holdings; a gap larger than one month is treated as a full exit (`hold_prev` = 0 in that gap).

- **Questionnaire structure: v1 splits employer versus employee respondents.** `v1` classifies the respondent ("我是小微经营者" selects the employer segment); it determines which subsequent questions the respondent is asked. Several indicators have two parallel question versions — one asked of employers, one asked of employees.

- **Platform hard constraints.** Reading and writing data is limited to the four functions listed below.

  - `ant_read_data`
  - `ant_print_all`
  - `ant_plot`
  - `ant_write_data`

  Output is capped at 50 KB per code block and 500 KB per file. `describe` / `corr` / `value_counts` require at least 100 input rows; for smaller tables, use `describe_by_discrete` on the large-table aggregate or `unique` on a Series. Figures are bounded to 3–12 inches and 50–150 DPI. `try-except` is not permitted.

2. 2026.9.1

2.1 Current SME-owner reduced form

This section summarizes the real-data platform run of the SME-owner reduced-form script, `01_sme_expectations_rf.py`.

- The run uses `ANSWER_SECONDS_MIN = 120`: respondents with observed answer time below 120 seconds are excluded, while missing answer time is retained.
- The sample is restricted to SME owners with positive beginning portfolio holdings and nonmissing passive returns.
- The analysis is reduced form: it asks whether predetermined passive portfolio returns move expectations.
- The key empirical result is a broad null. None of the 11 expectation outcomes is significant at the 5% level; `exp_price` is the closest result ($p = 0.0774$).

Specification.

$$Y_{iw} = \beta \cdot \texttt{shock_ret_rate_pp}_{iw} + \alpha_w + \gamma_{c(iw)} + \boldsymbol{x}'_{iw}\delta + \varepsilon_{iw}.$$

- Y_{iw} is one of 11 SME-owner expectation outcomes.
- `shock_ret_rate_pp` is the passive mutual-fund return in percentage points.
- α_w are wave fixed effects.
- $\gamma_{c(iw)}$ are portfolio-cell fixed effects.
- $\boldsymbol{x}_{iw}$ contains four predetermined traits: age, college indicator, firm age, and company-registration indicator. [add `female` and `employment size` as well.]
- Standard errors use HC1.

Source tables and survey construction.

- The run reads five SME survey waves [2024q2 to 2025q2], one 2024q2 add-on table, one fund-level shock panel, one historical-control panel, and fund holding snapshots.
- The 2024q2 addon supplies `exp_gdp`, `exp_house`, `exp_cpi`, and `exp_rate`.
- There is no 2024q2 addon counterpart for `exp_stock`; hence `exp_stock` is estimated on four waves.

Table 2.1. Input tables used in the real-data platform run.

Source	Wave / scope	Rows	Columns	Role in the analysis
Survey	2024q2	14,876	16	SME-owner screen, traits, and 2024q2 micro expectations
Survey	2024q3	12,975	21	SME-owner screen, traits, and expectation outcomes
Survey	2024q4	12,509	21	SME-owner screen, traits, and expectation outcomes
Survey	2025q1	9,056	21	SME-owner screen, traits, and expectation outcomes
Survey	2025q2	9,593	21	SME-owner screen, traits, and expectation outcomes
Addon	2024q2	14,876	5	Macro expectations missing from the main 2024q2 employer segment
Shock panel	all waves	122,359	4	Fund-level passive-return shock by wave
Control panel	all waves	1,266,578	4	Fund-month historical returns for risk and expected-return controls
Holdings	all waves	1,288,708	6	Beginning portfolio weights for shock and control aggregation

Table 2.2. Survey construction summary. All rows have nonmissing answer time in the platform output; the last column applies the 120-second cutoff.

Wave	Respondents	SME owners	SME-owner share	Answer $<$ 120s
2024q2	14,876	9,354	62.9%	4,399
2024q3	12,975	7,502	57.8%	2,729
2024q4	12,509	8,007	64.0%	3,030
2025q1	9,056	5,324	58.8%	2,455
2025q2	9,593	5,966	62.2%	2,512

- The 2024q2 addon merge is exact on the available keys: both survey and addon have 14,874 nonmissing `submit_id` keys, and the overlap is 14,874.
- The addon contributes 2,977 nonmissing observations for `exp_gdp`, 3,227 for `exp_cpi`, 485 for `exp_house`, and 705 for `exp_rate`.

2.2 CSMAR shock and control inputs

The fund-level inputs are built outside the Ant platform from the CSMAR fund files.

- The merge key is `fund_code` × `wave`. `fund_code` is the six-digit CSMAR `TradeCode`, normalized to match the Ant holding-table field 基金代码.[3]

Shock construction. For fund j and wave w, the CSMAR script computes a short-window return from cumulative NAV:

$$r_{jw}^{shock} = \frac{\text{AccNAV}_{j,\text{end}(w)}}{\text{AccNAV}_{j,\text{baseline}(w)}} - 1.$$

[3]Normalization is constructed by treating it as a string, stripping whitespace, removing any trailing `.0`, and left-padding numeric codes to six digits strings.

- The baseline is the latest valid cumulative NAV on or before the day before the shock window starts. [i.e., end of last month]
- The end value is the latest valid cumulative NAV inside the wave-specific shock window.
- The uploaded variable `shock_ret_rate_pp` equals $100 \times r_{jw}^{shock}$.
- A fund is usable when return is nonmissing, the return is based on cumulative NAV, the window is sufficiently complete[4], and the return is not extreme.
- Missing or unusable fund returns are not recoded as zero.

Table 2.3. CSMAR shock windows and fund-level usable rates.

Wave	Window start	Window end	History window	Funds	Usable share
24Q2	2024-06-01	2024-06-13	2023-06–2024-05	23,020	84.16%
24Q3	2024-09-01	2024-09-17	2023-09–2024-08	23,686	84.15%
24Q4	2024-12-01	2024-12-10	2023-12–2024-11	24,391	84.38%
25Q1	2025-03-01	2025-03-17	2024-03–2025-02	25,217	84.12%
25Q2	2025-06-01	2025-06-12	2024-06–2025-05	26,045	84.07%

Trading-day and quality filters.

- Trading days are inferred from CSMAR daily NAV coverage, rather than from a generic business-day calendar.
- A date is treated as a trading day when at least 1,000 funds report nonmissing cumulative NAV and the count reaches at least 20% of the maximum daily disclosure count. [That's an important technical issue. Let's discuss.]
- This rule drops holidays (e.g., Mid-Autumn Festival) and weekend dates with only a small number of special fund records.
- Completeness is defined as valid NAV dates divided by inferred trading days in the window.
- The current threshold is `complete_ratio >= 0.80`.
- The extreme-return flag is `|shock_ret_rate| > 0.20`; almost all funds pass this filter in the current output.

[4]Details mentioned below in "Trading-day".

User-level passive shock. After upload, the Ant-platform script merges `shock_panel.csv` to beginning holdings and constructs

$$\texttt{shock_ret_rate_pp}_{iw} = \frac{\sum_j H_{ij,w-1}\,\texttt{shock_ret_rate_pp}_{jw} \cdot \mathbf{1}\{\texttt{shock_usable}_{jw} = 1\}}{\sum_j H_{ij,w-1}}.$$

[if not unsale, should this fund be excluded from denominator as well?] The denominator is the beginning portfolio size. The script also reports return coverage, so incomplete shock coverage can be diagnosed separately from the realized value of `shock_ret_rate_pp`.

Control construction. The control input keeps monthly fund returns rather than pre-aggregated fund-level means and standard deviations:

$$r^H_{ijmw} = \frac{H_{ij,w-1} \cdot \texttt{ctrl_monthly_ret}_{jmw}}{\sum_k H_{ik,w-1}}.$$

The platform script first sums these weighted fund returns within each user × historical month to obtain a fixed-portfolio monthly return. It then computes:

$$\texttt{expected_ret_12m}_{iw} = \frac{1}{12}\sum_m R^H_{imw}, \qquad \texttt{portfolio_risk_12m}_{iw} = \mathrm{sd}_m(R^H_{imw}).$$

This is the Xiaohan-note portfolio-control 口径: risk is the standard deviation of the fixed portfolio's historical monthly return series, not a weighted average of individual fund risks.

Shock and control aggregation.

- ~~Holdings are taken from the end-of-previous-window snapshot specified in the script.~~ Holdings are measured at the month-end before each shock window (e.g., 2024-08 month-end holdings for the 2024q3 shock window).
- Fund-level shocks are aggregated with holding amounts as weights.
- Historical controls use the monthly control panel and the same beginning-portfolio weights.
- Return coverage is essentially complete in every wave; control coverage is high but not literally complete for all users.

Table 2.4. User-level passive-return and historical-control coverage.

Wave	Users	Mean port.	Ret. cov.	Ctrl. cov.	Min ctrl. cov.	Ctrl. months	R nonmiss	Risk nonmiss
2024q2	24,163	28,962.74	0.9996	0.9942	0.9852	11.37	22,890	22,905
2024q3	23,168	29,946.61	0.9996	0.9954	0.9867	11.35	21,904	21,933
2024q4	21,913	32,034.86	0.9996	0.9938	0.9851	11.45	20,896	20,925
2025q1	22,253	31,587.94	0.9997	0.9951	0.9877	11.09	20,557	20,583
2025q2	20,997	33,902.31	0.9995	0.9958	0.9904	11.41	19,957	19,986

Base sample and portfolio cells.

- The regression base sample imposes SME owner, answer-time cutoff, positive beginning portfolio, and nonmissing passive return.
- The platform output also reports stricter diagnostics for full return coverage, full 12-month control coverage, and nonmissing portfolio-cell inputs.
- Most sample attrition comes from the SME-owner screen, the answer-time cutoff, and the positive-portfolio requirement. [These are fair. Just need to discuss whether we should include zero-portfolio respondents.]

Table 2.5. Sequential sample funnel.

Step	2024q2	2024q3	2024q4	2025q1	2025q2
Raw survey rows	14,876	12,975	12,509	9,056	9,593
SME owners	9,354	7,502	8,007	5,324	5,966
After answer-time filter	8,093	7,059	7,402	4,942	5,455
Positive portfolio	4,770	4,086	3,760	2,686	2,976
R nonmissing	4,770	4,078	3,752	2,682	2,968
Return complete diag.	4,761	4,060	3,704	2,648	2,927
Control complete diag.	4,471	3,780	3,446	2,405	2,681
Cell inputs diag.	4,471	3,780	3,446	2,405	2,681

- Portfolio cells use 10 size bins, 5 historical-risk bins, and 5 historical-expected-return bins.
- In the real run, 18,245 user-wave observations have a constructed cell after removing 3 duplicate user-wave rows.
- There are 223 cells; only one is a singleton, and the largest cell has 272 observations.

2.3 Reduced-form results

- The main coefficient β is the response of each expectation outcome to a one-percentage-point passive portfolio return shock.
- All outcomes include wave fixed effects and portfolio-cell fixed effects. [add city and industry FEs and employment size, female, positive profit.]
- `exp_stock` uses 4 waves; all other outcomes use 5 waves.
- Across outcome-specific nonmissing samples, the number of portfolio cells ranges from 205 to 222.

Table 2.6. Main reduced-form results: coefficient on `shock_ret_rate_pp`.

Outcome	Obs.	$\hat{\beta}$	SE	t	p	R^2
`exp_stock`	3,290	-0.18208	0.22442	-0.811	0.4172	0.1640
`exp_gdp`	5,017	-0.03192	0.03929	-0.812	0.4165	0.0463
`exp_cpi`	5,419	-0.00763	0.02776	-0.275	0.7835	0.0731
`exp_house`	3,795	0.18392	0.17652	1.042	0.2974	0.0676
`exp_rate`	4,226	0.01090	0.01767	0.617	0.5373	0.4021
`exp_env_local`	7,429	0.00024	0.00657	0.036	0.9712	0.0454
`exp_rev`	12,858	-0.00049	0.05757	-0.009	0.9931	0.0218
`exp_market`	12,478	0.01020	0.05822	0.175	0.8609	0.0197
`exp_price`	5,714	0.05536	0.03134	1.766	0.0774	0.0462
`exp_wage`	6,169	0.02825	0.02243	1.260	0.2077	0.0445
`exp_input_cost`	5,923	-0.00947	0.02750	-0.344	0.7307	0.0584

Interpretation.

- The reduced form does not show robust evidence that passive stock-fund return shocks move SME-owner expectations.
- None of the 11 coefficients is significant at the 5% level.
- The closest result is `exp_price`: a one-percentage-point passive return shock is associated with a 0.055 percentage-point higher expected product-price change ($p = 0.0774$).
- This `exp_price` result is suggestive at the 10% level but should be treated as weak evidence, especially given the number of outcomes.
- Macro-expectation signs are mixed: stock, GDP, and CPI are negative; housing prices and interest rates are positive.
- Own-business expectation signs are also mixed and close to zero: local environment and revenue are essentially zero; market demand, product prices, and wages are positive; input costs are slightly negative.

To-do 2026.09.02

- Zero-portfolio respondents: no drop; set as 0 for passive gains/return (0911: set as baseline)
- Try `passive gains`
- Try drop 2024q3
- Specification: add `female`, `employment size`, city FE, and ind FE.
- $shock_ret_rate_pp_{iw}$ equation
- Don't pool all waves to create cells. Keep constructing by wave(month) and adding wave × cell FEs.

- Refer to the CSMAR document to see how to calculate `return`
- Balance test
- (the last thing I want to do) Match each fund's returns through the day before the survey submission date.

Controls. The trait coefficients mainly serve as balance controls rather than objects of interest.

- Age is positively related to `exp_stock` and `exp_cpi`.
- Age is negatively related to `exp_rev`, `exp_market`, and `exp_price`.
- College education is negatively related to `exp_house` and `exp_env_local`, but positively related to `exp_rate`, `exp_price`, `exp_wage`, and `exp_input_cost`.
- Firm age is negatively related to `exp_house`, `exp_env_local`, and `exp_rev`.
- Company registration is positively related to `exp_rev` and negatively related to `exp_env_local`.
- These associations are descriptive conditional correlations inside the reduced-form specification.

3. 2026.9.10

3.1 Old fine-cell runs

- Old design: passive return; positive portfolios; `10 × 5 × 5` size/risk/expected-return cells.
- Result: broad null; the marginal `exp_price` coefficient is not stable across runs.
- Implementation issue: continuous employee count reduces 12,169 rows to 5,678 and should not define the baseline sample.

3.2 Problems found by diagnostics

- Cell sparsity: `10 × 5 × 5` becomes thin in outcome-specific samples; `exp_stock` has 512 cells for 2,745 observations and 153 singletons.
- Variable handling: employment size is categorical; employer-industry fields require the corrected wave-specific mapping.
- Identification diagnostics: residual shock variation remains; observed balance failures are small.

3.3 The used specification choice

- Shock: usable passive return in percentage points; positive portfolios only.
- Cells: `5 × 3 × 3` size/risk/expected-return bins interacted with wave.
- Fixed effects: cell, city level, gender, employer industry, grouped employment size.
- Controls: age, college, firm age, company registration; historical-control completeness enforced.
- Inference: HC1 standard errors.

3.4 Results already run

- Analysis base: 14,385; outcome-specific samples: 2,510–9,687.

Table 3.1. Main reduced-form results under the post-diagnostic `5 × 3 × 3` specification.

Outcome	Obs.	$\hat{\beta}$	SE	t	p	R^2
`exp_stock`	2,510	-0.14748	0.28798	-0.512	0.6086	0.1813
`exp_gdp`	3,726	-0.06199	0.05086	-1.219	0.2229	0.0759
`exp_cpi`	4,064	0.01840	0.03683	0.500	0.6174	0.0951
`exp_house`	2,962	0.15003	0.21272	0.705	0.4806	0.0747
`exp_rate`	3,300	0.00296	0.02321	0.128	0.8983	0.4149
`exp_env_local`	5,740	0.00196	0.00903	0.217	0.8284	0.0599
`exp_rev`	9,687	0.05330	0.07951	0.670	0.5026	0.0375
`exp_market`	9,415	0.07113	0.07894	0.901	0.3675	0.0384
`exp_price`	4,451	0.03615	0.03967	0.911	0.3622	0.0723
`exp_wage`	4,827	0.02935	0.02495	1.176	0.2394	0.0722
`exp_input_cost`	4,595	-0.03431	0.03554	-0.966	0.3343	0.0698

3.5 Interpretation and balance checks

- Main result: **broad null**; no coefficient significant at 10%.
- Cell quality: substantially improved under `5 × 3 × 3`; meaningful within-cell and residual shock variation.
- Balance: broadly acceptable; significant trait predictors are economically small.
- Next step: retain the coarser design and compare return versus gain estimands.

4. 2026.9.20

Shock-definition comparison under the common coarser design. Unless noted otherwise: five waves; zero portfolios retained; `5 × 3 × 3` portfolio cells; control completeness enforced; HC1 standard errors.

4.1 Portfolio-cell construction

> The cell-FE should have been constructed **within wave**, not pooled across waves. All the runs below used pooled ranks, which is a accidental mistake and thus has to be rerun.

- Intended design: size, risk, and expected-return ranks computed **separately within each wave**; wave interacted with the resulting portfolio cell.
- Previous implementation: ranks computed once in the **pooled** multi-wave sample; wave label attached only after ranking.
- Previous zero treatment: recorded-zero and unmatched-holding observations filled with zeros; `rank(method="first")` then split tied zeros across bins according to row order (hence they were aligned **randomly**).
- Consequence: dropping one wave changed bin cutoffs and cell assignments in every retained wave, which caused the numerical explosion in the run that dropped `2024q3`.
- Corrected implementation: positive portfolios ranked within wave using equal-value-preserving average ranks; recorded-zero and unmatched/imputed-zero portfolios assigned to separate wave-specific zero cells and excluded from quantile ranking.

4.2 Run matrix and configuration checks

- Fixed effects: wave-specific portfolio cell, city level, gender, employer industry, grouped employment size.
- Controls: age, college, firm age, company registration.
- Main specification: `cell_other_fe_controls`.
- Configuration warning: the four-wave passive-return run omits `2024q3`.

Table 4.1. Shock definitions compared on 2026.09.20.

Run	Core regressor	Waves	Status	Interpretation
R1	`passive_return`	Five	Rerun	Pre-correction return/information result
R2	`passive_return`	Four	**Invalid**	Old cells; severe numerical degeneration
R3	`passive_gain`	Five	Rerun	Pre-correction wealth-shock result
R4	`realized_return`	Five	Rerun	Pre-correction descriptive result
R5	`realized_gain`	Five	Rerun	Pre-correction descriptive result

Table 4.2. Why the four-wave passive-return run is excluded: comparison with the same-unit R1.

Diagnostic	R2: four-wave passive return	R1: five-wave passive return
Included waves	2024q2, 2024q4, 2025q1, 2025q2	All five waves
Analysis base	20,795	26,694
Main-specification N	2,617–12,858	4,336–17,482
Absolute coefficient range	4.53–766.62	0.00349–0.08973
Standard-error range	120.40–985,814.86	0.00820–0.24788
R^2 pattern	All extremely negative; minimum below -10^8	Positive; 0.0173–0.3584
Reading	**Numerically degenerate; unusable**	Stable, but based on old pooled-rank cells

Invalid four-wave run.

4.3 Estimations across shock definitions

Table 4.3. Estimations across shock definitions.

Outcome	Obs.	Passive return	Passive gain	Realized return	Realized gain
exp_stock	4,336	0.08973	-0.00022	-0.00041***	-0.00015
		(0.24788)	(0.00020)	(0.00015)	(0.00011)
exp_gdp	6,777	-0.06893	0.00000	-0.00020***	-0.00001
		(0.04613)	(0.00003)	(0.00007)	(0.00002)
exp_cpi	7,343	0.01369	0.00004**	0.00002	0.00003*
		(0.03394)	(0.00002)	(0.00002)	(0.00002)
exp_house	5,494	0.07219	-0.00005	0.00013	-0.00008
		(0.18810)	(0.00019)	(0.00010)	(0.00013)
exp_rate	6,104	0.00569	0.00001	-0.00005***	0.00001
		(0.01924)	(0.00001)	(0.00001)	(0.00001)
exp_env_local	10,746	0.00349	-0.00000	-0.00003*	0.00000
		(0.00820)	(0.00001)	(0.00002)	(0.00000)
exp_rev	17,482	0.08629	0.00000	0.00020***	-0.00001
		(0.07315)	(0.00004)	(0.00004)	(0.00003)
exp_market	16,995	0.08310	-0.00006	0.00001	-0.00003
		(0.07174)	(0.00005)	(0.00002)	(0.00004)
exp_price	8,154	0.03987	-0.00004	-0.00001	-0.00003
		(0.03706)	(0.00003)	(0.00002)	(0.00002)
exp_wage	8,888	0.01786	-0.00004*	-0.00001	-0.00002*
		(0.02306)	(0.00002)	(0.00001)	(0.00001)
exp_input_cost	8,514	-0.02703	-0.00004**	0.00001	-0.00002
		(0.03150)	(0.00002)	(0.00004)	(0.00002)

Notes: HC1 standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

- Passive return: broad null under old pooled-rank cells.
- Passive gain: positive CPI; negative input cost; negative wage at 10% under old pooled-rank cells.
- Multiple testing: no passive-gain coefficient survives a strict correction across 11 outcomes.
- Realized return/gain.

4.4 R3 across specifications

Table 4.4. Passive-gain coefficients across R3 specifications.

Outcome	Cell FE	All FE	All FE + controls
exp_stock	-0.00001	-0.00003	-0.00022
	(0.00016)	(0.00016)	(0.00020)
exp_gdp	-0.00001	-0.00000	0.00000
	(0.00002)	(0.00002)	(0.00003)
exp_cpi	0.00005***	0.00004***	0.00004**
	(0.00002)	(0.00002)	(0.00002)
exp_house	-0.00007	-0.00008	-0.00005
	(0.00016)	(0.00016)	(0.00019)
exp_rate	0.00000	0.00000	0.00001
	(0.00001)	(0.00001)	(0.00001)
exp_env_local	-0.00000	0.00000	-0.00000
	(0.00001)	(0.00001)	(0.00001)
exp_rev	-0.00001	-0.00001	0.00000
	(0.00004)	(0.00004)	(0.00004)
exp_market	-0.00006	-0.00006	-0.00006
	(0.00005)	(0.00004)	(0.00005)
exp_price	-0.00004*	-0.00004*	-0.00004
	(0.00003)	(0.00002)	(0.00003)
exp_wage	-0.00004**	-0.00004**	-0.00004*
	(0.00002)	(0.00002)	(0.00002)
exp_input_cost	-0.00003	-0.00003*	-0.00004**
	(0.00002)	(0.00002)	(0.00002)

Notes: HC1 standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

- CPI: positive and significant in all three specifications; modest attenuation after controls.
- Wage: negative and stable in magnitude; precision weakens after controls.
- Input cost: negative coefficient strengthens as FE and controls are added.
- Product price: negative at 10% without controls; loses significance after controls.
- Other outcomes: no stable evidence across specifications.

4.5 Passive-gain sample composition

Table 4.5. Portfolio provenance in the five-wave passive-gain analysis base.

Portfolio status	Observations	Share of base
Recorded positive portfolio	14,385	53.9%
Recorded zero portfolio	882	3.3%
No holding record, imputed zero	11,427	42.8%
Total analysis base	26,694	100.0%

- **Zero exposure is the main unresolved sample issue.** A total of 12,309 observations, or 46.1% of the base, enter with zero passive gain.

- Most zero-gain observations are not verified zero portfolios: 11,427 have no matched holding record and are imputed to zero.
- Fund-shock coverage is otherwise high across waves. Mean usable portfolio size rises from about 28,799 yuan in 2024q2 to 33,591 yuan in 2025q2.

4.6 Passive-gain cell and variation diagnostics

Table 4.6. Ranges of outcome-specific passive-gain diagnostics.

Diagnostic	Minimum	Maximum	Reading
Occupied cells	67	90	Modest cell count under `5 × 3 × 3`
Singleton cells	0	4	Very limited singleton problem
Share of observations in cells with $n < 5$	0.08%	0.86%	Little observation-weighted sparsity
Median cell size	20.0	70.5	Adequate central cell density
Share in cells with no gain variation	30.7%	38.1%	Large structural-zero component
Raw gain SD, yuan	1,242.7	1,470.4	Substantial unconditional variation
Within-cell gain SD, yuan	1,149.3	1,387.9	Substantial within-cell variation
Residual gain SD, yuan	1,147.0	1,386.1	FE and controls retain variation
Residual sum-of-squares share	80.7%	89.4%	Limited absorption by FE and controls

- Density: limited sparsity in the outcome samples.
- Residual variation: more than 80% of raw passive-gain sum of squares remains after residualization.
- Structural-zero signal: 31–38% of observations lie in cells with no gain variation, consistent with the old zero/imputed-zero treatment.

Table 4.7. Failed passive-gain lottery-balance checks at the 5% level.

Outcome	Trait	p	Standardized effect	Obs.
`exp_price`	Firm age	0.0206	0.0229	9,496
`exp_rev`	Company registration	0.0341	-0.0169	21,072
`exp_rate`	Firm age	0.0484	0.0223	7,359

- There are three failures among roughly 55 outcome-trait tests; about 2.75 false positives are expected mechanically at a 5% threshold.
- All standardized effects are close to two percent of a shock standard deviation. There is no concentrated failure across outcomes or traits.

4.7 Working interpretation and next benchmark

- **Working main estimand: passive gain.** This aligns the project with a yuan wealth-shock interpretation and with the quasi-lottery gain design in Andersen et al.
- **Current status: rerun required.** No R1–R5 coefficient should be treated as final under corrected cells.
- Keep-zero concern: 42.8% of the analysis base consists of unmatched holding records previously imputed to zero.
- Immediate benchmark: five-wave passive gain; corrected within-wave cells; positive portfolios only.
- Keep-zero robustness: recorded-zero and imputed-zero observations placed in distinct zero cells; neither group enters positive-portfolio quantile ranking.
- Passive return should remain a complementary information/return-shock specification. Realized return and realized gain remain descriptive endogenous comparisons.

5. 2026.10.29

5.1 Current specification

The exposure is passive gain in RMB 10,000. The full-control regression is

$$Y_{iw} = \beta \cdot \texttt{passive_gain}_{iw} + \gamma_{c(iw)} + \boldsymbol{x}'_{iw}\delta + f_w(P_{iw}) + \varepsilon_{iw}.$$

Notation.

- i, j and w index owners, funds and survey waves. Y_{iw} is an expectation outcome; $\gamma_{c(iw)}$ denotes wave-by-portfolio-cell effects and therefore absorbs the earlier α_w.
- $\boldsymbol{x}_{iw}$ contains the selected personal/business controls. P_{iw} contains initial portfolio amount, historical risk and historical mean return. The owner index is retained because these characteristics differ within wave.

Passive gain and the return reconstruction.

- **Aggregation.** Let A^0_{ijw} be the initial holding amount in yuan and R_{jw} the fund's shock-window return in decimal units. The implemented exposure is

  $$\texttt{passive_gain}_{iw} = \frac{1}{10{,}000} \sum_{j \in \mathcal{U}_{iw}} A^0_{ijw} R_{jw},$$

  where $\mathcal{U}_{iw}$ contains holdings with usable shock returns under the current coverage rules. Holdings are fixed at the initial snapshot; subsequent trades do not enter. This is an amount, not a return rate or realized trading profit.
- **Why reconstruction was needed.** Earlier CSMAR monthly returns contained spurious zeros for money-market funds. Ordinary-fund cumulative-NAV growth also required review: cumulative and unit NAV need not use the same share units after past distributions or conversions. These issues affect both the shock and the twelve-month portfolio controls.
- **Money-market funds.** Affected monthly returns were replaced using disclosed income per share under the accepted normalization and within-month compounding rule. Income-based shocks compound $(1 + d_{j\tau}/10{,}000)$ over disclosure dates inside the shock window, minus one; $d_{j\tau}$ is the disclosed income amount under that convention. Bundled weekend/holiday income crossing month-end belongs entirely to its disclosure month; it is not split across months. Income recorded after the shock end is not added. Unresolved or invalid income records remain missing. This is a posted-income convention, not exact daily accrual or a verified fund-specific settlement model; some field/par-value combinations still need validation.

- **Ordinary funds: intermediate versus current rule.** An intermediate candidate used $(\text{AccNAV}_{\text{end}} - \text{AccNAV}_{\text{start}})/\text{NAV}_{\text{start}}$. External checks showed that a simple denominator replacement is not valid for all cumulative-NAV conventions. The current inputs instead use $R = \text{NAV}_{\text{end}}/\text{NAV}_{\text{start}} - 1$. Calculable price returns are retained with event/quality flags; the optional screen is off. These are not distribution- or split-adjusted total returns. Money-fund values preserve the accepted reconstruction rather than being replaced by ordinary-fund price returns.

- **Historical characteristics.** Using fixed initial portfolio weights, reconstruct each of the twelve historical monthly portfolio returns. Their arithmetic mean μ_{iw} and sample standard deviation σ_{iw} enter the cells and continuous controls. Risk includes cross-fund covariance; it is not an average of fund volatilities. These controls use the reconstructed monthly series, including the money-fund disclosure-month rule.

Wave-specific portfolio controls: $f_w(P_{iw})$.

- **Cells and continuous controls.** Cells and q_{iw} use the same three underlying characteristics: initial amount, historical risk and mean return. Cells use within-wave bins; q_{iw} retains continuous values after transformation. Thus f_w adjusts residual within-cell differences, allowing nonlinear and wave-specific relationships. It supplements coarse grouping rather than merely seeking a balance-test pass.

- **Construction.** For positive portfolios, define A_{iw} as portfolio amount under the analysis scope and

$$q_{iw} = \big(\log A_{iw},\ 100\sigma_{iw},\ 100\mu_{iw}\big)', \qquad z_{k,iw} = \frac{q_{k,iw} - \bar{q}_k}{s_k}.$$

  The full-control adjustment is

$$f_w(P_{iw}) = a_w' q_{iw} + \sum_{k=1}^{3}\sum_{\ell=k}^{3} b_{k\ell,w} z_{k,iw} z_{\ell,iw} + d_w \frac{A_{iw}}{10{,}000}.$$

  There are three linear terms, three squares, three pairwise interactions and one amount-level term per wave. The linear terms use q; the polynomial terms use standardized z.

- **Why amount levels also enter.** Gain equals amount times portfolio return. If $R_{iw} = \bar{R}_w + \Delta R_{iw}$, then its common-return component is $(A_{iw}/10{,}000)\bar{R}_w$. Amount interacted with wave can absorb this component; log amount alone cannot do so exactly. This can also remove substantial identifying variation.

- **Implementation.** Each term is multiplied by a wave indicator and entered jointly in OLS. Means $\bar{q}_k$ and scales s_k are fixed using the five-wave eligible positive-portfolio reference, before

outcome-specific filtering. They and cell assignments are not recomputed after deletions. Zero-holder design terms are set to zero within separate zero cells; this does not assign a literal zero historical risk or return. For positive holders, invalid inputs remain missing.

- **Independent balance evidence.** OLS residual gains are mechanically orthogonal to included control columns in the fitted sample. Testing those same columns again is not independent evidence of balance. Informative checks instead examine strictly pre-shock traits or expectations not directly absorbed by the adjustment, on the relevant diagnostic sample. Failure to detect such associations supports the conditional comparison but does not establish random assignment of passive gains.

Current settings.

- **Sample and cells.** Within-wave $10\times3\times3$ amount/risk/mean-return cells; separate zero-holding groups; twelve-month history. The 2024q2 macro supplement remains included; `exp_stock` starts in 2024q3.
- **Timing and inference.** Submission must follow the shock-window end. Observed answer times below 180 seconds are excluded; missing durations are retained. Standard errors cluster by owner.
- **Exposure coverage.** Current inputs are `nav_price_annotated_v1`; quality flags do not trigger automatic exclusion. Recorded-zero and unmatched-as-zero holdings remain distinct groups. Positive-holder estimates provide a separate check.

Control specifications.

- **Broad-sample cells / Cell only.** Cell effects on the broad eligible sample / the complete-case sample.
- **Owner controls.** Cell effects plus age, college, gender, city tier, firm age, registration and broad industry.
- **Portfolio controls / Full controls.** Add the linear and polynomial terms in f_w / additionally add the amount-level term. All complete-case specifications use the same observations within each outcome.

5.2 Outcome definitions

Table 5.1. Outcome definitions and units.

Code	Chinese name	Meaning / forecast horizon	Scale
exp_cpi	CPI 涨幅预期	National CPI inflation.	%
exp_gdp	GDP 增速预期	National GDP growth.	%
exp_house	房价涨幅预期	National house-price growth.	%
exp_rate	贷款利率预期	Loan rate; later waves ask the level in 12 months for maturities above five years.	%
exp_stock	上证指数预期	Shanghai Composite level in 12 months.	Index points
exp_env_local	当地营商环境预期	Local business environment over the next year.	−1, 0, +1
exp_rev	营业收入预期	Next-quarter sales revenue growth.	%
exp_market	市场需求预期	Next-quarter demand / new orders growth.	%
exp_price	产品或服务价格预期	Next-quarter output-price growth.	%
exp_wage	全职员工平均工资预期	Next-quarter growth in average wages paid to full-time staff.	%
exp_input_cost	原材料价格预期	Next-quarter raw-material price growth.	%

Notes: Percentage intervals are assigned representative values; their regression coefficients are in percentage points. The environment score is −1 (worse), 0 (unchanged), +1 (better). Later-wave macro growth forecasts cover twelve months; firm outcomes compare next quarter with the current quarter. The 2024q2 supplement follows the established mapping.

Raw-material prices are not total operating costs; average wages are not owner income or total payroll.

5.3 Main coefficients and interpretation

Table 5.2. Passive-gain coefficients across control specifications.

Outcome	Cell only	Owner controls	Full controls
Macro and local expectations			
`exp_stock` ($N = 4,346$)	−50.597	−57.643	−49.971
	(61.550)	(61.105)	(62.049)
`exp_gdp` ($N = 7,132$)	0.241	0.197	0.095
	(0.268)	(0.265)	(0.250)
`exp_cpi` ($N = 7,723$)	0.397**	0.390**	0.490**
	(0.187)	(0.187)	(0.199)
`exp_house` ($N = 5,573$)	−1.780	−1.854	−2.189
	(2.285)	(2.238)	(2.486)
`exp_rate` ($N = 6,204$)	0.135	0.147	0.189
	(0.107)	(0.109)	(0.138)
`exp_env_local` ($N = 11,252$)	0.048	0.043	0.076
	(0.074)	(0.074)	(0.077)
Firm expectations			
`exp_rev` ($N = 18,252$)	0.059	0.049	0.136
	(0.438)	(0.450)	(0.535)
`exp_market` ($N = 17,750$)	−0.557	−0.592	−0.448
	(0.522)	(0.531)	(0.620)
`exp_price` ($N = 8,490$)	−0.404	−0.374	0.069
	(0.292)	(0.289)	(0.314)
`exp_wage` ($N = 9,291$)	−0.389*	−0.391*	−0.395**
	(0.219)	(0.217)	(0.185)
`exp_input_cost` ($N = 8,895$)	−0.583***	−0.585***	−0.310*
	(0.205)	(0.208)	(0.178)

Notes: User-clustered standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (two-sided, unadjusted). Each entry is the passive-gain coefficient from a separate outcome regression; gain is in RMB 10,000. The complete-case sample is fixed across columns within each outcome. Owner controls add personal/business characteristics; full controls also add wave-specific portfolio terms and portfolio amounts.

- **`exp_cpi`.** Cell-only and owner-control estimates are close (0.397 and 0.390); the full-control estimate is 0.490 (SE 0.199, $p = 0.014$). Thus RMB 10,000 more passive gain is associated with 0.49 percentage points higher coded `exp_cpi`.

- **`exp_wage`.** Owner-control and full-control estimates are similar (-0.391 and -0.395; full controls $p = 0.033$). The coefficient concerns expected growth in average full-time pay.

- **`exp_input_cost`.** The coefficient attenuates from -0.585 with owner controls to -0.310 with full controls ($p = 0.082$), a 47% reduction in magnitude.

- **Other outcomes.** The remaining eight full-control estimates provide no clear overall response; imprecision does not imply zero effects.

- **Sample selection.** Broad-to-common N falls from 9,016 to 7,723 for `exp_cpi` and from 10,328 to 8,895 for `exp_input_cost`. Complete-case comparisons hold N fixed. Full-control positive-holder estimates are close to baseline: 0.481 for `exp_cpi` and -0.295 for `exp_input_cost`.

5.4 Actual deletion refits

Table 5.3. Sensitivity of passive-gain coefficients: full controls.

Tail deletions					
Outcome	Baseline	Capital 0.5%	Capital 1%	Residual gain 0.5%	Residual gain 1%
`exp_cpi`	0.490**	0.541	0.714	0.749	0.721
	(0.199)	(0.466)	(0.555)	(0.644)	(0.788)
`exp_wage`	−0.395**	−0.413	−0.251	−0.966	−1.089*
	(0.185)	(0.405)	(0.539)	(0.649)	(0.659)
`exp_input_cost`	−0.310*	0.021	−0.241	0.816	0.690
	(0.178)	(0.459)	(0.587)	(0.675)	(0.761)
Wave omissions					
Outcome	Omit 2024q2	Omit 2024q3	Omit 2024q4	Omit 2025q1	Omit 2025q2
`exp_cpi`	0.754**	0.375*	0.486**	0.484**	0.462**
	(0.323)	(0.213)	(0.199)	(0.201)	(0.208)
`exp_wage`	−0.531	−0.308*	−0.386**	−0.366*	−0.436**
	(0.324)	(0.170)	(0.184)	(0.197)	(0.196)
`exp_input_cost`	−0.380	−0.263*	−0.302*	−0.356**	−0.276
	(0.337)	(0.156)	(0.178)	(0.179)	(0.193)

Notes: User-clustered standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (two-sided, unadjusted). Each entry is the passive-gain coefficient from a separate outcome regression; gain is in RMB 10,000. Columns identify the sample deletion; all regressions use full controls.

Capital tails use within-wave initial-amount cutoffs, retaining boundary ties. Residual tails rank squared baseline residual gains; deletion counts use positive holders. Each scenario refits controls and fixed effects while keeping cell assignments and scaling fixed.

- **Tail sensitivity.** Full-control `exp_cpi` remains positive (0.541–0.749), but SE rises to 0.466–0.788 and all $p > 0.19$. Owner-control `exp_cpi` changes sign after the 1% residual deletion (-0.131, SE 0.687). Full-control `exp_input_cost` estimates change sign across tail rules; `exp_wage` estimates remain negative with owner or full controls.

- **User influence.** Separately deleting the two highest absolute-score users gives full-control `exp_cpi` 0.631 ($p = 0.002$) and 0.410 ($p = 0.033$). Scores depend on the outcome: these are diagnostics, not sample-selection rules. Both deletions retain positive `exp_cpi` and negative `exp_input_cost`.

- **Wave omissions.** The coefficient signs for `exp_cpi`, `exp_wage` and `exp_input_cost` survive every omission with owner or full controls. Removing 2024q3 lowers full-control `exp_cpi` to 0.375 ($p = 0.078$) and owner-control `exp_cpi` to 0.195. `exp_stock` has no q2 observations, so its q2 omission adds no evidence.

5.5 Balance, concentration and wave heterogeneity

Table 5.4. Identification diagnostics and their limits.

Check	Result	Interpretation / limitation
Prior investor traits	4 of 110 tests have p<0.05.	Selected repeat respondents; no q2 prior coverage.
Own prior expectations	22 checks; N=161–1,702. `exp_house`: $p = 0.008$ with owner controls; $p = 0.604$ with full controls.	Earlier answer is a placebo outcome, not a same-target forecast revision.
Top 1% residual SS share	Full-control `exp_cpi` 81.92%; `exp_input_cost` 82.90%.	A small set of positive-holder observations supplies most variation.
Largest observation SS share	Full-control `exp_cpi` 21.65%; `exp_input_cost` 23.69%.	Observation influence need not equal whole-user influence.
q2+q3 residual SS share	Full-control `exp_cpi` 81.88%; `exp_input_cost` 79.18%.	Contribution shares are not wave-specific causal slopes.
After 1% residual deletion	Remaining SS: `exp_cpi` 9.67%; `exp_input_cost` 9.32% of baseline.	Lower concentration comes with a large information loss.

Balance checks use strictly pre-shock age, college, firm age, registration and employment, with cell or portfolio adjustment. Four tests have $p < 0.05$: age in the `exp_rev`/`exp_market` samples and firm age in the `exp_market` sample. Prior expectations use the previous scheduled response, dated before the current shock; no backfill. Limited historical coverage prevents a general balance claim.

Table 5.5. Equality of wave-specific gain slopes: unadjusted Wald p-values.

Outcome	Owner controls	Full controls
`exp_stock`	0.000002	0.117086
`exp_gdp`	0.000422	0.155219
`exp_cpi`	< 0.000001	< 0.000001
`exp_house`	< 0.000001	0.000007
`exp_rate`	0.000074	0.001825
`exp_env_local`	0.723476	0.398896
`exp_rev`	0.595164	0.750254
`exp_market`	0.571960	0.232250
`exp_price`	0.171343	0.143267
`exp_wage`	0.353597	0.003018
`exp_input_cost`	< 0.000001	0.005469

The equality tests indicate wave heterogeneity. In 2024q4, full-control `exp_cpi` is -47.768 (SE 6.365) and `exp_input_cost` -41.771 (SE 11.197). Yet omitting q4 barely changes pooled estimates (0.490 to 0.486; -0.310 to -0.302). These interaction slopes need support and influence checks before economic

interpretation.

5.6 Conclusions and limitations

- **Working conclusion.** `exp_cpi` is a positive exploratory signal; `exp_wage` estimates show a consistent negative direction. `exp_input_cost` estimates are more control- and tail-sensitive. The evidence remains conditional association, not a general causal wealth effect.
- **Identification.** Portfolio selection and business shocks may jointly affect gains and expectations. Current traits are not all verified pre-shock. Historical balance checks cover selected repeat respondents.
- **Support.** Tail deletion removes about nine tenths of identifying variation for `exp_cpi` and `exp_input_cost`. A large N therefore does not establish broad support. Wave heterogeneity further limits a common-effect interpretation.
- **Measurement and inference.** Interval coding and inapplicable answers affect outcomes and coverage. NAV-price changes may differ from total returns. Unmatched holdings need not imply zero wealth, and user clustering leaves common-fund dependence unresolved.
- **Next checks.** Audit answer mappings and q4 residual support; consider strictly pre-shock business-scale/profitability balance. Operating-cost and employment expectations are possible extensions after coding checks.

Sources: verified core, tail and wave outputs (10281700, 10291050, 10291330); definitions follow *小微调研变量映射表.xlsx*. All 22 repeated baseline estimates match. Tables report rounded estimates; no new regressions were run.